RealPDE Track 2 (Long-Term Test-Time Adaptation): final submission in brief

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Task. Streaming prediction of real PIV airfoil flow: at each step the model sees 20 frames of (u, v, p) on a 32×64 grid, returns the next 20 frames plus per-element interval bounds, and then receives the true frames. Batch size 1, strict trajectory order, every step timed.

Model (trained offline, only on the released data).

- **Backbone:** a SimVP-style gated spatiotemporal-attention CNN (3.4 M parameters) with causal context normalisation (per-pixel running mean/std of all frames revealed so far, fed as extra inputs and used to de-standardise the output). Robust Welsch loss.
- **Sim pretraining** on all 100 simulation files (10k iterations), then **real fine-tuning** on 81 real files (8k iterations).
- **Weight soup:** 4 fine-tunes from the same pretrained snapshot are averaged in weight space.
- **Ensemble:** 3 such soups from 3 different pretrain seeds, packed into one batched network (10.2 M parameters) with an fp16 core.
- **Auxiliary nets:** a 4-seed energy net (fluctuation amplitude) and an interval-width net fitted on out-of-sample residuals.

Test-time adaptation (causal, gradient-free, per trajectory).

- **Least-squares correction (LSK):** a ridge regression, updated online from the revealed truth, maps 8 SVD directions of the ensemble's head features to a correction of the u, v output (forgetting factor 0.97).
- **Energy calibration:** the energy net rescales the predicted fluctuations; the blend with the revealed energy history is chosen per trajectory from the previous window's score.
- **Low-energy damping:** fluctuations are shrunk where the prediction's own kinetic energy is low.
- **Intervals:** width = net output × running error scale × an adaptive multiplier; centre shifted toward the raw ensemble output by an adaptive weight. Both are chosen per trajectory by scoring a small grid on the previous window with the exact interval-score formula.
- **Regime-change reset:** a detector on the input free-stream velocity restarts the running statistics (including the least-squares moments) after an abrupt change.

Execution. The whole step (with separate graphs for the first two windows) runs as CUDA graph replays with bit-exact bit-tested fallbacks. Work independent of the backbone runs on parallel streams, and elementwise chains are fused into runtime-compiled kernels with exact rounding. Outputs are returned by value across the harness's process boundary. On the eval harness, a step takes 4.84 ms.

Result (development leaderboard, 27 Sep 2026).

rel_l2	TKE	MVPE	time	SPS	**final**
94.939	81.017	94.500	92.464	45.211	**82.452**

First on the development-phase leaderboard (Codabench submission 948041). After the organisers retrained the top methods for the final evaluation, this submission won Track 2.

Reproducibility. One command, `bash scripts/retrain_all.sh` (run in `final_submission/realpde/` of the code release), rebuilds the submission from the released data in about 2–3 h on an RTX 3090 or 2 h 15 on an A100 with the evaluation software. Seeds are fixed and `train_real/7575_0.h5` is excluded throughout.